Samuel Zarr

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Education

University of Wisconsin—Madison 2023 - 2027

Computer Sciences BS, Cellular and Molecular Biology BS - GPA 3.9

- Honors: Deans list, Phi Beta Kappa
- Relevant Coursework: Honors Algorithms, AI in Genomics, Cell Biology

Research

Student Researcher – Yichiao Zhong Lab | Madison, WI 8/25-present

- Studying how transformer-based biology foundation models learn and store cellular dynamics.
- Implemented linear probing and Sparse Autoencoders (SAE) to extract internal information from biology models.
- Performed statistical analysis comparing model predictions and experimental data to validate model interpretability.

Research Intern – Gerber Lab at Harvard Medical School | Boston, MA 6/23-9/24

- Led the processing and statistical analysis of terabytes of genomic data
- Conducted a literature search for novel statistical and computational methods for taxonomic reconstruction
- Applied a high-performance computing cluster to test various data processing techniques
- Prepared data for probabilistic deep learning models
- Benchmarked ML topic models learning relations within multiomic data

Research Intern – French Lab at Harvard Medical School | Boston, MA 6/21-9/22

- Investigated pathogenesis of NUT Carcinoma using wet lab techniques.
- Investigated lncRNA expression and trialed RNA degraders in cell culture.
- Analyzed genomic screening data using R to identify molecular targets in NUT carcinoma.
- Used Python and R to design and generate high-dimensional data visualizations for publication.

Clubs and Activities

Wisconsin AI Safety Initiative | UW Madison 2023-present

- Leader coordinating large-scale education and upskilling programs, university relations, professional networking, and outreach for organization with 130+ members and $30,000 budget
- Organizer of the 2026 AIMS symposium, bringing together distinguished professors, elected officials, and AI researchers to discuss how to build a better society

Skills

Languages: `Python (Numpy, Pandas), R (DESeq2), Bash, C++, Java`

Tools & Infrastructure: Git, Docker, High-Performance Computing (HPC), PyTorch

Wet Lab: Cell culture, RNAscope, FISH, Western blotting, Confocal microscopy